Cisco Devices Management

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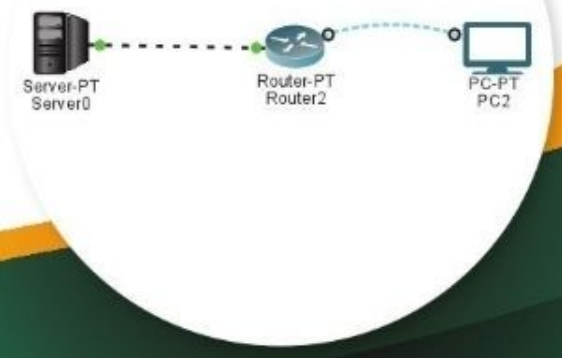
The knowledge of the components used in a router and switch is extremely useful while learning how to manage a Cisco internetwork. Understanding the configuration of Cisco devices helps you manage them in a better way. The IOS files and the configuration files of a Cisco device are stored in different locations. In this chapter, you will learn about the configuration register and how to use the configuration register for password recovery. This chapter also provides information about how to upgrade or back up the Cisco IOS from a TFTP server. You will also learn about licensing of IOS devices.

OBJECTIVES

By the end of this chapter, you will be able to:

- Understand maintenance of configuration register
- Describe password recovery at configuration register
- Describe how to upgrade or back up the Cisco IOS from a TFTP server and to a TFTP server
- Understand licensing of IOS devices



Configuration Register Maintenance at Router

The Cisco router has a 16-bit software register. It is written in NVRAM. It is set to load the IOS from flash and load the startup-config file from the NVRAM.

CERTIFICATION OBJECTIVE

Understand maintenance of configuration register
Describe password recovery at configuration register

The 16 bits of configuration register are read from left to right (15 to 0). By default, bits 13, 8, and 1 are set to 1, as shown in Table 2, because the default configuration value is 0x2102. Each set of 4 bits is called a nibble and read in binary and has values 8, 4, 2, 1.

Configuration Register value	2				1				0				2			
Bit number	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Binary number	0	0	1	0	0	0	0	1	0	0	0	0	0	0	1	0

Table 1. Default Configuration Register Bit numbers

IMPORTANT NOTES

0x at the configuration register shows that the following digits are in hexadecimal format.

How to check the current configuration register value

`The show version` command is used to check the current value of configuration register. An example using this command is shown below. The last line of output displays the value of the configuration register.

```
Router#show version
Cisco Internetwork Operating System Software
IOS (tm) PT1000 Software (PT1000-I-M), Version 12.2(28),
RELEASE SOFTWARE (fc5)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2005 by cisco Systems, Invc.
Compiled Wed 27-Apr-04 19:01 by miwang
Image text-base: 0x8000808C, data-base: 0x80A1FECC
ROM: System Bootstrap, Version 12.1(3r)T2, RELEASE SOFTWARE
(fc1)
```

```
Copyright (c) 2000 by cisco Systems, Inc.
ROM: PT1000 Software (PT1000-I-M), Version 12.2(28), RELEASE
SOFTWARE (fc5)

System returned to ROM by reload
System image file is "flash:pt1000-i-mz.122-28.bin"

PT 1001 (PTSC2005) processor (revision 0x200) with 60416K/5120K
bytes of memory
.

Processor board ID PT0123 (0123)
PT2005 processor: part number 0, mask 01
Bridging software.
X.25 software, Version 3.0.0.
4 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)
Configuration register is 0x2102
```

Example 1: Current value of configuration register

Changing the Configuration Register

The configuration register value can be changed to modify how a router boots and runs. The main reasons for changing the configuration register values are:

- To force the system to ROM monitor mode.
- To change the default boot source and boot filename.
- To control broadcast addresses.
- To load IOS from ROM
- To boot from TFTF server

IMPORTANT NOTES

show version displays system hardware information, serial number, software version and other information.

`The config-register` command is used to change the configuration. Example 2 demonstrates how to change the configuration and then display the new configuration value.

```
Router(config)#config-register 0x2101
Router(config)#^Z
Router#show ver
Cisco Internetwork Operating System Software
IOS (tm) PT1000 Software (PT1000-I-M), Version 12.2(28),
RELEASE SOFTWARE (fc5)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2005 by cisco Systems, Inc.
Compiled Wed 27-Apr-04 19:01 by miwang
Image text-base: 0x8000808C, data-base: 0x80A1FECC
[OUTPUT CUT]
2 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)
Configuration register is 0x2102 (will be 0x2101 at next reload)
```

Example 2: Changing configuration register value

IMPORTANT NOTES

0x2142 bypasses the startup-config and the router directly goes to setup mode.

When the router is uploaded, the configuration register value will be automatically changed. Example 3 shows the updated register value.

```
Router>en
Router#sh ver
Cisco Internetwork Operating System Software
IOS (tm) PT1000 Software (PT1000-I-M), Version 12.2(28),
RELEASE SOFTWARE (fc5)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2005 by cisco Systems, Inc.
Compiled Wed 27-Apr-04 19:01 by miwang
Image text-base: 0x8000808C, data-base: 0x80A1FECC
[OUTPUT CUT]
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)
Configuration register is 0x2101
```

Example 3: Changed Register Value

Recovering Passwords

If you forgot the password of your router and are not able to access it, then by changing the value of the configuration register you can regain the password to your router. Bit 6 in the configuration register is used to recover the password. But, in the default configuration, bit 6 is set to 0. To recover the password, bit 6 should be set to 1. This is done by setting the configuration value to 0x2142.

To recover a password, complete the following steps:

1. Boot the router and interrupt the boot sequence to go into the ROM monitor mode. This can be done by the `break` command.
2. Change the configuration register value to 0x2142.
3. Reload the router.
4. Enter the Privilege EXEC mode.
5. Copy the startup-config file to the running-config file.
6. Change the password.
7. Set the register value back to the default value.

Step 1: Boot the router and interrupt the boot sequence

You can interrupt the boot sequence by pressing Ctrl+Break key. The last line shows that you are in the ROM monitor mode. The output will be as shown in Example 4.

```
System Bootstrap, Version 12.1(3r)T2, RELEASE SOFTWARE (fc1)
Copyright (c) 2000 by cisco Systems, Inc.
Initialising memory for ECC
..
c2811 processor with 524288 Kbytes of main memory
Main memory is configured to 64 bit mode with ECC enabled

Readonly ROMMON initialised

Self decompressing the image :
####
monitor: command "boot" aborted due to user interrupt
rommon 1 >
```

Example 4: Interrupting boot sequence

Step 2: Change the configuration value to 0x2142 using the `config-register` command. Use the command `config 0x2142` for the ISR/2600 router.

Step 3: Reset the router by using the `reset command`. The router will reload and ask if you want to enter the setup mode. Answer no and enter the User EXEC mode, then type `enable` to enter the Privilege EXEC Mode.

Step 4: Copy the startup-config file to the running-config file by using the command:

```
copy startup-config running-config
```

Or by the using the shortcut command:

```
copy start run
```

IMPORTANT NOTES

0x2142 bypasses the startup-config and the router directly goes to setup mode.

Now, the configuration is changed. Enter the configuration mode to set the password by using the commands:

```
configure terminal
enable secret abcdef
```

Step 5: Reset the value of the configuration register to default.

ANSWER IT

1. Which keystroke interrupts the boot sequence on a Cisco router to implement password recovery procedures?
   a. Ctrl-Alt-Delete
   b. Ctrl-Break
   c. Fn-F4
   d. Fn-Break
2. What is the default configuration register on most Cisco routers?
   a. 0x2142
   b. 0x2100
   c. 0x2102
   d. 0x2202
3. When you need to recover a password, what bit number do you manipulate in the configuration register?
   a. 1
   b. 4
   c. 6
   d. 8

Backing Up and Restoring the Cisco IOS

CERTIFICATION OBJECTIVE

Describe how to upgrade or back up the Cisco IOS from a TFTP server and to a TFTP server

Before you try to back up an IOS image on a TFTP server, you must:

- Confirm that you can access the TFTP server.
- Confirm that the server has adequate storage space for the IOS image.
- Verify the file naming and path requirements.

If a server's Ethernet port is directly connected to a router's Ethernet interface, as shown in Figure 1, then you need to verify the following:

- Administrator's server must have the working TFTP server software.
- The router's and the server's Ethernet must be connected via a crossover cable.
- The subnet of the server's Ethernet and the router's Ethernet interface should be in the same subnet.

Figure 1: Backing up the IOS

Taking Backup of the Cisco IOS

`The copy flash tftp` command is used to back up the Cisco IOS of your router to the TFTP server. To ensure that there is a connectivity between the router and the TFTP server, ping the TFTP device from your router.

Your output will be as shown in Example 5.

```
Router#ping 192.168.2.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.2, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
0/0/3 ms

Router#
```

Example 5: Pinging the TFTP server

After pinging, you will get a confirmation that the connection to the server is working, then use the `copy flash tftp` command to copy the IOS from flash to TFTP server as shown in Example 6. Copy the IOS filename from either the `show flash` or `show version`, and paste the name when asked for source file name.

```
Router#copy flash tftp
Source filename []? pt1000-i-mz.122-28.bin
Address or name of remote host []? 192.168.2.2
Destination filename [pt1000-i-mz.122-28.bin]?

Writing pt1000-i-mz.122-28.bin....!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!
[OK - 5571584 bytes]

5571584 bytes copied in 3.572 secs (357395 bytes/sec)
```

Example 6: Copying the IOS file from flash to TFTP server

Restoring or Upgrading the IOS

If you need to restore or upgrade the IOS of your router, then you can use the `copy tftp flash` command to upgrade or restore the router by copying the IOS image from the TFTP server. Before upgrading the IOS, ensure that the flash has enough space to store the IOS. Use commands shown in

Example 7 to upgrade or restore the IOS.

```
Router#copy tftp flash
Address or name of remote host []? 192.168.2.2
Source filename []? pt1000-i-mz.122-28.bin
Destination filename [pt1000-i-mz.122-28.bin]?
%Warning:There is a file already existing with this name
Do you want to over write? [confirm]

Accessing tftp://192.168.2.2/pt1000-i-mz.122-28.bin...
Loading pt1000-i-mz.122-28.bin from 192.168.2.2: !!!!!!!!!!!!
!!!!!!!!!!!
```

The technology packages are:

- IP Base: This is a prerequisite for installing other features. It provides entry-level IOS functionality.
- Data: MPLS, ATM, and multiprotocol support
- Unified Communications: VoIP and IP telephony
- Security: Cisco IOS Firewall, IPS, IPSec, 3DES, and VPN

The technology packages can be activated manually or by using an application called the CLM (Cisco License Manager). This free tool runs on Windows and communicates with the Cisco product license registration portal on the Internet to install license keys.

Those routers that support the new licensing model have a unique device identifier (UDI). It is a combination of the product ID (PID) and a serial number (SN). The `show license udi` command is used to show this information as shown in Example 8.

```
Router#show license udi
Device#   PID              SN              UDI
-------------------------------------------------------
-------------------------------------------------------
*1   CISCO1941/K9 FTX1524Z0ZW  CISCO1941/K9:FTX1524Z0ZW
```

Example 8: License UDI

When you purchase the software package, it comes with a PAK (Product Authorisation Key). PAK and UDI are used to activate the license. The PAK has a unique number and Cisco uses it to check the license details. When you enter your details on the CLM, Cisco will e-mail you the license key. The `license install` command is used to install the license key.

The `show license all` command is used to view all the licenses installed on a device. It also displays various variables related to software licensing and activation.

Example 9 shows this command on a router.

```
Router#show license all
License Store: Primary License Storage
StoreIndex: 0 Feature: ipbasek9 Version: 1.0
        License Type: Permanent
        License State: Active, In Use
        License Count: Non-Counted
        License Priority: Medium
License Store: Evaluation License Storage
StoreIndex: 0 Feature: securityk9 Version: 1.0
        License Type: Evaluation
        License State: Inactive
             Evaluation total period: 208 weeks 2 days
             Evaluation period left: 208 weeks 2 days
        License Count: Non-Counted
        License Priority: None
StoreIndex: 1 Feature: datak9 Version: 1.0
        License Type:
        License State: Inactive
             Evaluation total period: 208 weeks 2 days
             Evaluation period left: 208 weeks 2 days
        License Count: Non-Counted
        License Priority: None
```

Example 9: Licenses on a router

The `show license feature` command also displays the details of licensed and unlicensed features in a summarised form. Example 10 shows this command on a router.

```
Router#show license feature
Feature name  Enforcement  Evaluation  Subscription  Enabled
ipbasek9      no           no          no            yes
securityk9    yes          yes         no            no
datak9        yes          no          no            no
```

Example 10: Show license feature command

The `show version` command also shows the license information. Example 11 shows this command on a router.

```
Router#show version
Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M),
Version 15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 23-Feb-11 14:19 by pt_team
[OUTPUT CUT]
License Info:

License UDI:
-------------------------------------------------
Device#        PID                 SN
-------------------------------------------------
*0             CISCO1941/K9        FTX1524Z0ZW

Technology Package License Information for Module:'c1900'
-------------------------------------------------------
-------------------------------------------------------
Technology     Technology-package          Technology-package
               Current Type                 Next reboot
-------------------------------------------------------
ipbase         ipbasek9        Permanent      ipbasek9
security       None            None           None
data           None            None           None
```

Example 11: License information at show version command

Recap

- The 16-bit configuration register in Cisco router loads the IOS from flash and the startup-config file from NVRAM.
- 0x2102 is the default value of the configuration register and 0x2142 is used for the password recovery of the router.
- The steps in the password recovery process interrupt the router boot sequence, change the configuration register, reload the router, and enter the Privilege EXEC mode, change/set the password, save the new configuration, reset the configuration register, and reload the router.
- The `copy flash tftp` command is used to back up the IOS image from flash memory to a TFTP server and to restore or upgrade the IOS image from TFTP server to a flash memory.
- The Cisco ISR G2 routes run the IP Base image by default, but extra features are unlocked by a license key.
- A CLM is used to activate the technology packages and get the license key.
- UDI is a combination of the PID and SN. The `show license udi` command is used to show UDI.
- `The show license all` command is used to view all the licenses installed on a device and `show license feature` command also shows the details in summarised form.

Test Your Skills

Multiple Choice Questions

1. What command would you use to configure the router to not load the startup configuration?
   a. `config-register 2102`
   b. `config-register 1200`
   c. `config-register 2100`
   d. `config-register 2142`
2. You wish to verify the version of the Cisco IOS you are running. What command would you use?
   a. `show ios`
   b. `show memory`
   c. `show version`
   d. `show software`
3. You wish to view the current value of the configuration register. What command would you use?
   a. `show reg`
   b. `show version`
   c. `show flash`
   d. `show config`
4. You wish to remove a license from your Cisco device. What command would you use?
   a. `remove license ipservices`
   b. `delete ipservices`
   c. `license clear ipservices`
   d. `delete license ipservices`

5. You wish to display the current license information. What command would you use?
   a. `display license file`
   b. `show license file`
   c. `license show file`
   d. `show all`
6. You need to back up your router configuration to a TFTP server. Which of the following commands would you use?
   a. `R1#copy tftp running-config`
   b. `R1#copy running-config tftp`
   c. `R1#backup running-config tftp`
   d. `R1#restore running-config tftp`

Scenario Based Questions

1. Using this output of the `show version` command, how many Fast Ethernet ports exist on the router?

```
Cisco 2811 (MPC860) processor
(revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
M860 processor: part number 0, mask 49
2 FastEthernet/IEEE 802.3 interface(s)
4 Low-speed serial(sync/async) network interface(s)
239K bytes of NVRAM
62720K bytes of processor board System flash (Read/
Write)
Configuration register is 0x2102
Router>
```